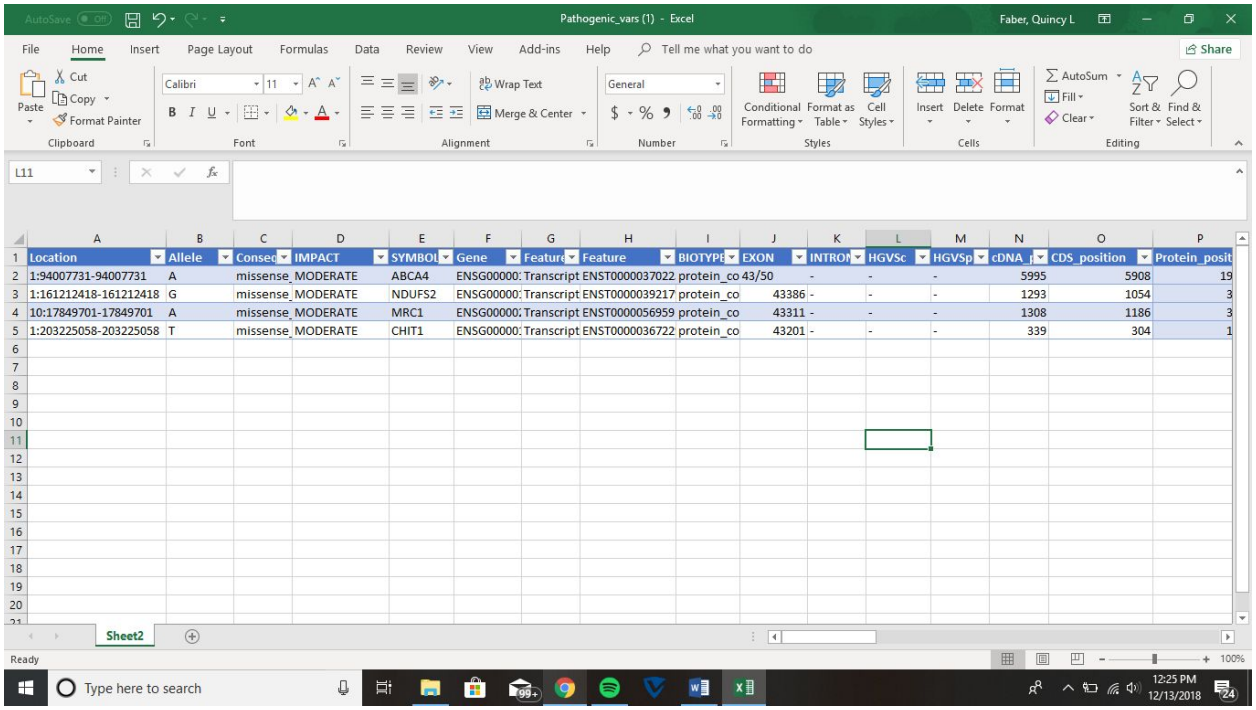
**Exome Chosen**: HG00284



| Location | Allele | Consec | IMPACT | SYMBOL | Gene | Feature | Feature | BIOTYPE | EXON | INTRON | HGVSc | HGVSp | cDNA_p | CDS_position | Protein_posit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1:94007731-94007731 | A | missense_ | MODERATE | ABCA4 | ENSG00000: | Transcript | ENST0000037022 | protein_co | 43/50 | - | - | - | 5995 | 5908 | 19 |
| 1:161212418-161212418 | G | missense_ | MODERATE | NDUFS2 | ENSG00000: | Transcript | ENST0000039217 | protein_co | 43386 | - | - | - | 1293 | 1054 | 3 |
| 10:17849701-17849701 | A | missense_ | MODERATE | MRC1 | ENSG00000: | Transcript | ENST0000056959 | protein_co | 43311 | - | - | - | 1308 | 1186 | 3 |
| 1:203225058-203225058 | T | missense_ | MODERATE | CHIT1 | ENSG00000: | Transcript | ENST0000036722 | protein_co | 43201 | - | - | - | 339 | 304 | 1 |

**Analytical Pipeline:**

We used the GRCh38 reference genome and prepared it for mapping through Samtools using the command: “bwa index <ref.fa>”. After our reference was prepared, we mapped our reads to it. We used the HG00284 exome as our sample. The command used to map our reads was: “`bwa mem -R '@RG\tID:foo\tSM:bar\tLB:library1' <ref.fa> <read1.fa> <read1.fa> > lane.sam`”. We then cleaned up the flags and read pairing information using: “`samtools fixmate -O bam <lane.sam> <lane_fixmate.bam>`”. Our BAM file was then converted to genomic positions using a pipe: “`bcftools mpileup -Ou -f <ref.fa> <sample1.bam> <sample2.bam> <sample3.bam> | bcftools call -vmO z -o <study.vcf.gz>`”.
We then fed our vcf into VEP to get our list of variants. The variants were then filtered to show only the missense variants.

**Table of Variants Called in Exome:**

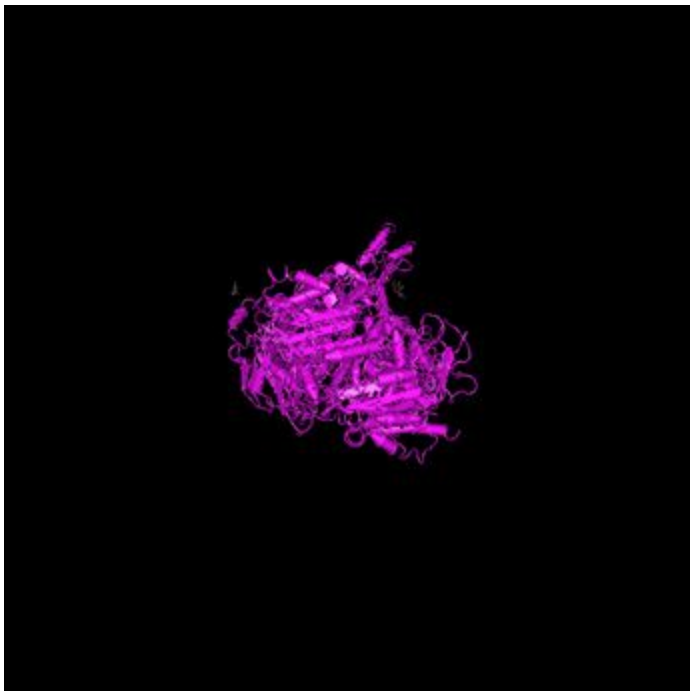
**Individual Variants**
**ABCA4 (Quincy Faber):**

Gene Name: ATP-binding cassette, sub-family A (ABC1), member 4 ABCA4
Location: 1:94007731-94007731
Nucleic Acid Change:Ttc to Ctc
Amino Acid Change: F to L
Variant: unknown

Known Variant: rs28938473 It is an A to G at 1:94473287.
It has 0.002925 allele frequency and 0.00384 in European (Non-Finnish).

This similar variant changes Leucine to Phenylalanine and is associated with Stargardt Disease. This disease causes degeneration of the macula, which is the center of the retina. The ATP-binding cassette protein normally works as an enzyme for photoreceptor cells.

Reference:
Rozet JM, et al. Spectrum of ABCR gene mutations in autosomal recessive macular dystrophies. *Eur J Hum Genet.* 1998 May-Jun;6(3):291-5.



Only ABCA1 has the wild-type protein structure solved. Its PDB ID is 5XJY.

**NDUFS2 (Jake Holt):**
Gene Name: NADH:ubiquinone oxidoreductase core subunit S2.
Location: 1:161212418-161212418
Nucleic Acid change: G to C
Amino acid change: A to P, Gct to Cct.
Variant: Unknown.
Known Variant:  rs11576415

The variant rs11576415 has a global allele frequency of 0.07613 (gnomAD) and a European allele frequency of 0.09817 (gnomAD).

NDUFS2 codes for a subunit of the mitochondrial Complex 1 protein. The known variant is associated with mitochondrial Complex 1 deficiency because it is a mutation in a gene coding for a subunit of Complex 1. However, there are no particular disease phenotypes that have been linked it, and it is considered to be likely benign. However, the mutation S314P (a different variant, that does not have an rs number) in the NDUFS2 gene is known to cause mitochondrial Complex 1 deficiency, and it has been linked to cardiomyopathy and encephalomyopathy in a mutational analysis study of patients with mitochondrial Complex 1 deficiency (Loeffen J, 2001). This mutation has not been published on gnomAD.

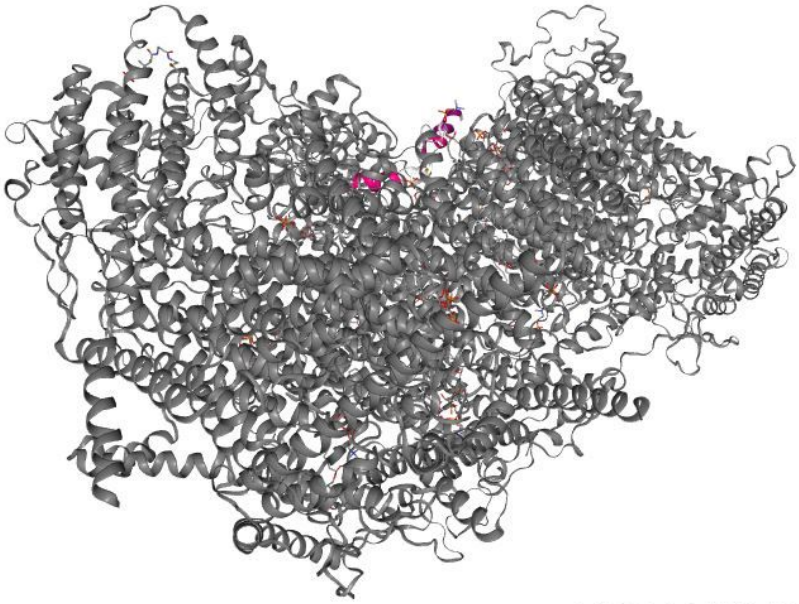
Since both the S314P mutation and the unknown variant result in a change from a non-cyclic amino acid to the cyclic proline amino acid, even though they are not in the same place, it is likely that they would have similar consequences. Proline could have a profound effect on how the protein folds because of its cyclic nature. Proline is known to inhibit rapid folding of proteins, and it's bulk could have other consequences as the protein folds. How mutations in the genes coding for subunits of the protein Complex 1 (which is needed as an enzyme to carry out oxidative phosphorylation in the mitochondria) cause disease phenotypes in patients is unknown because much is still unknown about how the protein is assembled from its subunits. It is widely accepted however that mutations in any of the genes coding for subunits of mitochondrial Complex 1 often result in mitochondrial Complex 1 deficiency. However, how that manifests clinically in patients can vary widely depending on which subunit is affected and which mutation is present.

Reference:

Loeffen J, Elpeleg O, Smeitink J, Smeets R, Stöckler-Ipsiroglu S, Mandel H, Sengers R, Trijbels F, van den Heuvel L. 2001. Mutations in the complex I *NDUFS2* gene are associated with hypertrophic cardiomyopathy and encephalomyopathy. *Ann Neurol* 49: 195–201.

Protein View:

The structure of the wild type or the mutant individual subunit has not been solved, but the complete wild type mitochondrial Complex I protein has been.



PDB ID 5XTC.Q

**CHIT1 (Taylor Cooper):**
Gene Name: Chitinase 1 (Chitotriosidase)
Variant ID: rs2297950
Location: 1:203225058-203225058
Nucleic Acid Change: C>T
Amino Acid [codon] Change: G[GGC] >S[AGC]
Variant frequency in the overall human population and in the particular ethnic/regional population:

Global: T=0.30532 (gnomAD)
European: T=0.30598 (gnomAD)

This gene encodes the enzyme chitotriosidase, which is secreted by human macrophages. As such, it's expression is heightened in bone marrow, lymph nodes, and lung tissue. The enzyme is highly conserved and is thought to play a role in degrading the cell walls of chitin-containing pathogens.

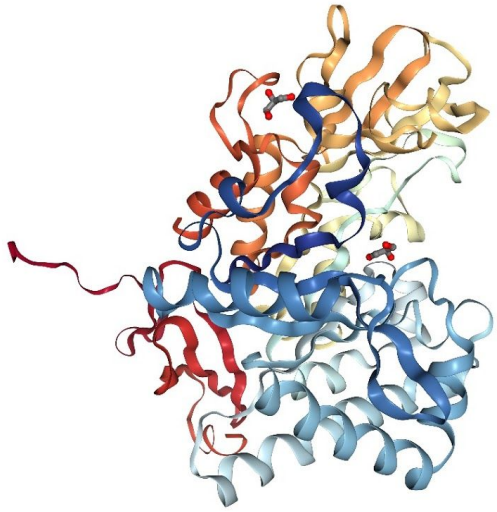
The rs2297950 SNP results in chitotriosidase deficiency and results in diminished defense against the chitin-containing pathogens. This SNP has also been seen in inflammatory

conditions, for example asthma and inflammatory bowel disease. However, chitotriosidase deficiency is typically a benign condition.

Because chitotriosidase functions as a chitin-cleaving enzyme and it predominantly resides in the lungs, it is important in relation to mold-related asthma. Fungi have chitinous cell walls, and increased fungal exposure is associated with severe asthma. Wu and colleagues concluded that three SNPs (rs2486953, rs4950936, rs1417149) in CHIT1 were associated with increased trips to the emergency room in families with high fungal exposure (2010). Because functional chitotriosidase breaks down chitin in the lungs, dysfunctional chitinase enzymes may lead to chitin accumulation and the development of allergies or asthma.

Reference:
Wu, A.C., J. Lasky-Su, C.A. Rogers, B.J. Klanderman, and A.A. Litonjua. 2010. Fungal Exposure Modulates the Effect of Polymorphisms of Chitinases on Emergency Department Visits and Hospitalizations. *American Journal of Respiratory and Critical Care Medicine*, 182: 884-889.

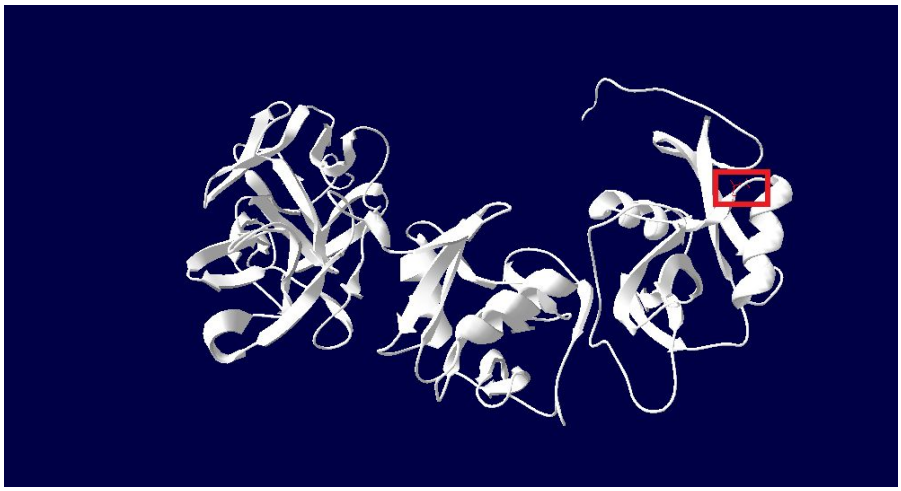
PDB ID: 5HBF
Only the wild-type protein structure of CHIT1 has been solved



**MRC1 (Josiah Davidson):**
Gene Name: MRC1 (Macrophage mannose receptor protein 1)
Location**:** 10:17849701-17849701
Variant: rs606231248, CM099897

This is a single point mutation from Guanine to Alanine which results in a missense codon converting Glycine to Serine
Overall Population frequency: 0.30955
European frequency: 0.00003

The variant has been published in a 2010 paper. This protein binds to the carbohydrates on cell walls, and even though this variant did not result in significant structural changes, it did affect the susceptibility to paucibacillary leprosy. Due to the insignificant structural changes, it is believed that there is a hindered interaction between a host molecule and the mutated valine residue. Only one photo is included below to show the structural mutation, but no significant structural change is observed.



Reference:
Alter, A., de Léséleuc, L., Van Thuc, N., Thai, V. H., Huong, N. T., Ba, N. N., Cardoso, C. C., Grant, A. V., Abel, L., Moraes, M. O., Alcaïs, A., … Schurr, E. (2009). Genetic and functional analysis of common MRC1 exon 7 polymorphisms in leprosy susceptibility. Human genetics, 127(3), 337-48.